

UNIT-II

Energy Storing Devices

LP 1: Relation between electrical energy and energy content of a cell: Batteries - Characteristics of battery- Reversible and irreversible cell

Introduction

Energy storage devices are systems which store energy in various forms such as electrochemical, kinetic, pressure, potential, electromagnetic, chemical, and thermal; using, for example, fuel cells, batteries, capacitors, flywheels, compressed air, pumped hydro, super magnets, hydrogen, etc.

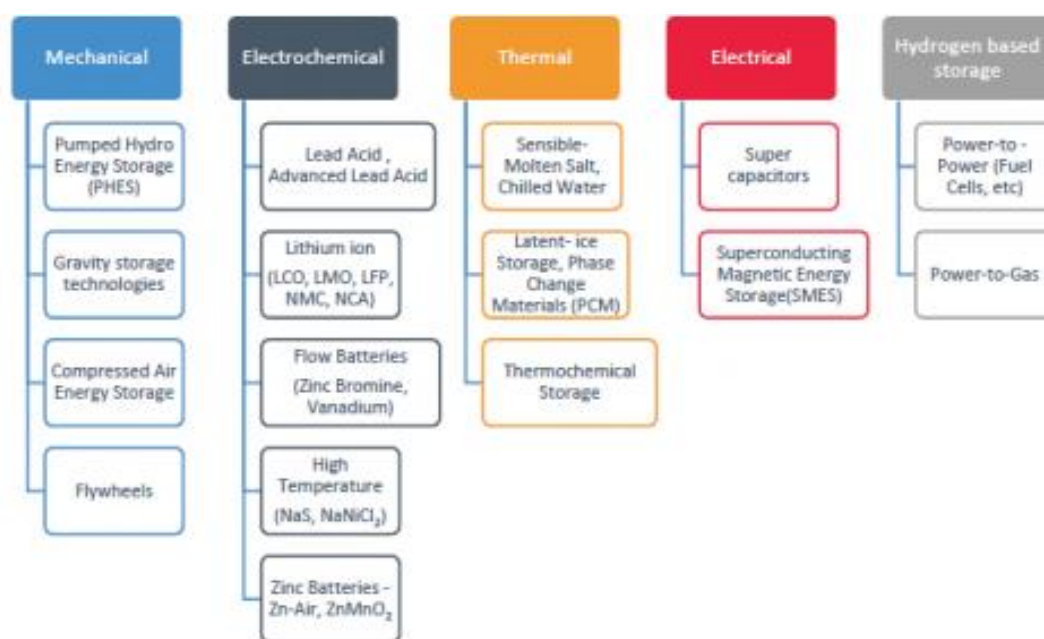


Figure 1: Classification of Energy Storage Technologies

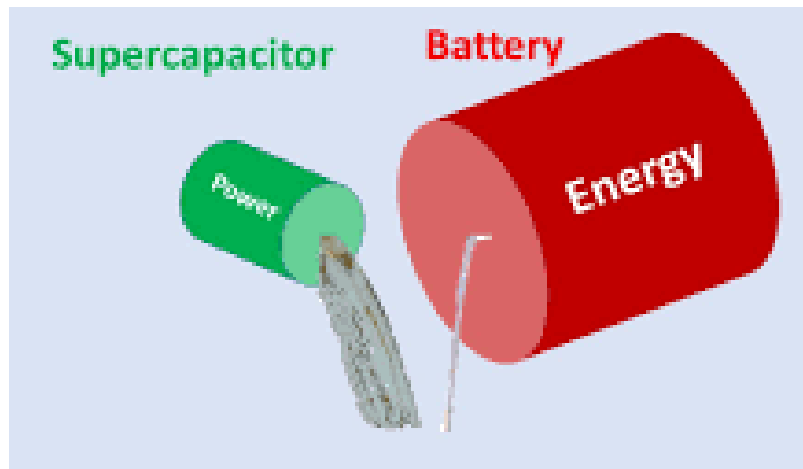
Electrochemical energy storing devices

Electrochemical energy storage is based on systems that can be used to view high energy density (batteries) or power density (electrochemical condensers). Current and near-future applications are increasingly required in which high energy and high power densities are required in the same material.

Electrochemical energy is what we normally call the conversion of chemical energy into electrical energy or vice versa. This includes reactions transferring electrons, redox reactions (reduction- oxidation). Reduction, when a substance receives one electron.

Energy density and power density of battery

They both measure an electrical characteristic of a battery in comparison to its weight. Nevertheless, energy density and power density differ in one important way: while battery energy density measures the watt hours (wh) per kilogram (kg), power density measures the watt output per kilogram.



Ragone plot

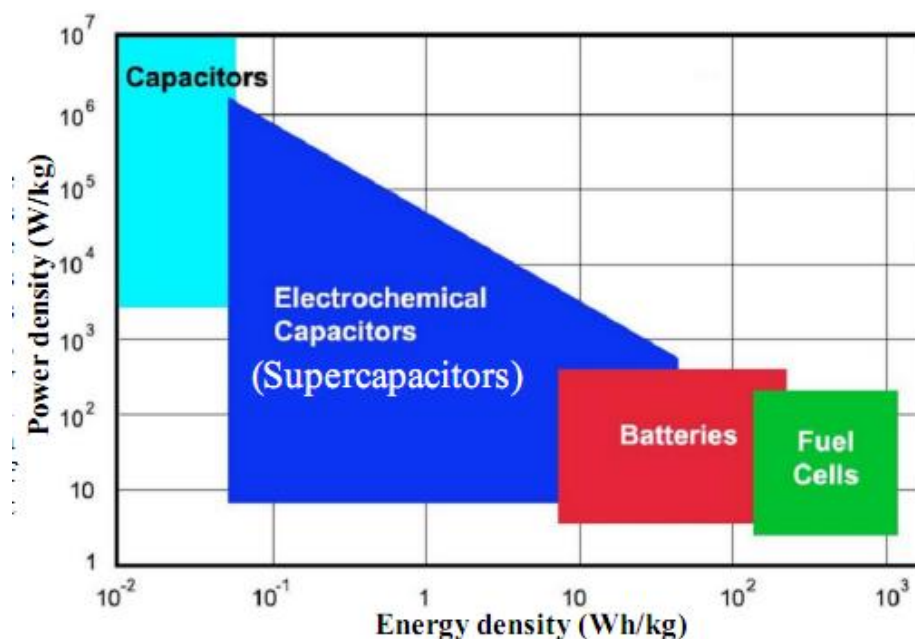


Fig.2. Power density vs. energy density of various energy storage systems.

- The performances of electrochemical capacitors can be compared in the Ragone chart plotting their respective energy and power densities as illustrated in Fig.2 for different electrical energy storage devices.

- Due to their physical charge storage, capacitors feature very large power densities compared with batteries and fuel cells but low energy densities. On the other hand, batteries and fuel cells have large energy densities but low power densities due to their slow reaction kinetics.
- Electrochemical capacitors bridge the gap between capacitors and batteries/fuel cells. They offer the prospect of maintaining the high energy density of batteries without compromising the high power density of capacitors.

History of battery

- The conversion of chemical energy into electrical energy is the basis for the functioning of a galvanic cell, named after its inventor Luigi Galvani (1786). “Voltaic piles” of unlike metals in contact with an electrolyte, developed by Alessandro Volta (1800) was the first power source.
- A major advance in the evolution of the battery was the Daniell cell invented by J.F. Daniell (1836).
- Plante (1859) developed the essentials of the lead-sulfuric acid storage cell.
- In 1868 George Leclanche introduced a cell which was the forerunner of the present day dry cell. Because of the chemical similarities, the dry cell, is still referred to as a Leclanche-type cell.
- The first true dry cell was developed between 1886 and 1888 by Dr. Carl Gassner. T.A. Edison, Adams, Ruben, are the few others who have contributed to the development of battery technology.

Cell & Battery

A cell designates a single unit. A battery is an arrangement of two or more cells, usually connected in series or parallel to supply the necessary current or voltage or both.

Difference Between Cell and Battery	
Cell	Battery
A cell is known to be a single unit device that tends to convert the chemical energy into electric energy.	A battery generally consists of a group of different cells.
Depending on what types of electrolytes are used, a cell can be either of reserve, wet or dry kinds. Cells can also include a molten salt type.	A battery can either be a primary or a secondary battery which means that it is either rechargeable or non-chargeable.
A cell is generally compact and light since it contains a single unit.	A battery consists of many different cells which give it a bigger size and make it bulkier.
A cell can only supply power for a shorter period.	A battery can supply power for much longer durations.
A cell is generally used for lighter tasks that do not require much energy. For example, it is used in clocks and lamps.	A battery is used generally for heavy-duty tasks. For example, it is used in automobiles and inverters.
Cells are cheaper in general.	Batteries are comparatively costlier.

Potential (voltage) – measured in volts. The open circuit voltage is defined by the chemistry (i.e., the active materials). It is independent of the size of the battery.

Current – measured in amps. This corresponds to the rate at which electrons can be removed from the battery. The current capability of a battery depends on the cell design and the chemistry.

Power – measured in watts. This is the product of the potential and the current: for a given current, the higher the voltage, the higher the power.

Capacity – usually measured in amp-hours. This is a measure of the number of electrons that can be removed from the battery. The capacity is proportional to the size of the battery.

All batteries contain:

- Anode - negative electrode
- A material that undergoes oxidation during the cell discharge.
- Cathode - positive electrode
- A material that undergoes reduction during the cell discharge.

- Electrolyte - medium for ion transfer
- A medium, usually liquid, through which ions move from one electrode to the other during the cell discharge. An ionic species, the electrolyte salt, is dissolved in the electrolyte.

Basic requirements and specification of commercial battery

- Portability
- Should be compact and light weight
- Should provide economically priced, continuous electric supply
- High cell voltage and stable voltage plateau over most of the discharge
- High stored energy content per unit mass and per unit volume
- Low cell resistance
- High sustained power output
- Wide temperature range of operation
- Long inactive shelf life and long operational life.
- Low initial cost
- Sealed and leak proof
- Rugged and resistant to abuse
- Safe in use and under accident conditions
- Should be capable of recharging with a capacity of many charge and discharge cycles, and to withstand overcharge and over discharge
- Sealed and maintenance free

Anode material should exhibit the following properties

- Efficient reducing agent
- High coulombic output
- Good conductivity
- Stable
- Ease of fabrication
- Low cost
- Metals such as Zinc and Lithium are often used as anode materials.

Cathode material should exhibit the following properties

- Efficient oxidizing agent.

- Stable when in contact with electrolyte
- Useful working voltage
- Ease of fabrication
- Low cost
- Metallic oxides such as are often used as cathode materials

The most desirable anode-cathode material combinations are those that result in light-weight cells with high voltage and capacity. Such combinations may not always be practical as a result of extenuating factors such as material handling difficulty, reactivity with other cell components, difficulty of fabrication, polarization tendencies, and cost prohibitive materials.

Electrolytes should exhibit the following properties

- Strong ionic conductivity
- No electric conductivity
- Non-reactivity with electrode materials
- Properties resistance to temperature fluctuations
- Safeness in handling
- Low cost
- Aqueous solutions such as dissolved salts, acids, and alkalis are often used as electrolytes

Working of a battery

- Batteries have three parts – an anode (-ve charge), a cathode (+ve charge) and an electrolyte.
- Cathode and anode are connected to an electrical circuit.
- Chemical reactions in the battery causes build up of electrons at the anode resulting in potential difference between the electrodes.

Characteristics of a battery

Size

- Several designations are used to mention the size of a battery. Example American National standards institute (ANSI), Japanese industrials standards (JIS), etc.,
- AAA, AA, C, D.

Lifetime or Longevity:

- For rechargeable batteries- as a result of fully charged battery lifetime decreases and the number of charge/discharge cycles.

- For non-rechargeable batteries- ‘shelf life’ is used to describe how long a battery will retain its performance between manufacture and use.
- Available capacity of all batteries drops with decreasing temperature.

Behavioral factors

- Temperature range (storage, operation)
 - As the **temperature** becomes lower, the chemical reaction slows down and a smaller current is produced but the **battery** is able to last longer.
 - As the **temperature** increases, the chemical reaction becomes faster and more current is produced but the **life** of the **battery** shortens
 - **If batteries** are exposed to excessive temperature, they will stop working, bulge, bubble, create sparks and flames, damage your device, or blowup. Extreme **heat** can lead to **battery** corrosion that shortens the average car **battery** life.

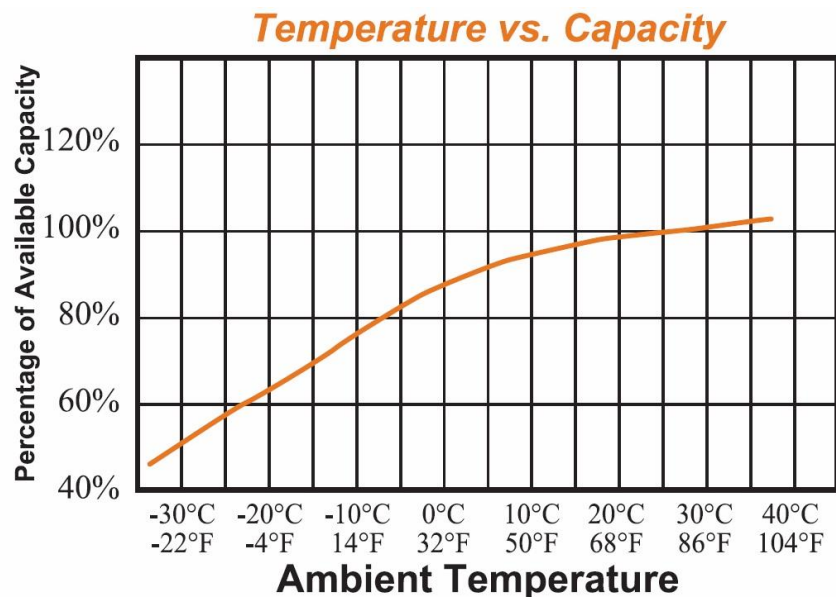
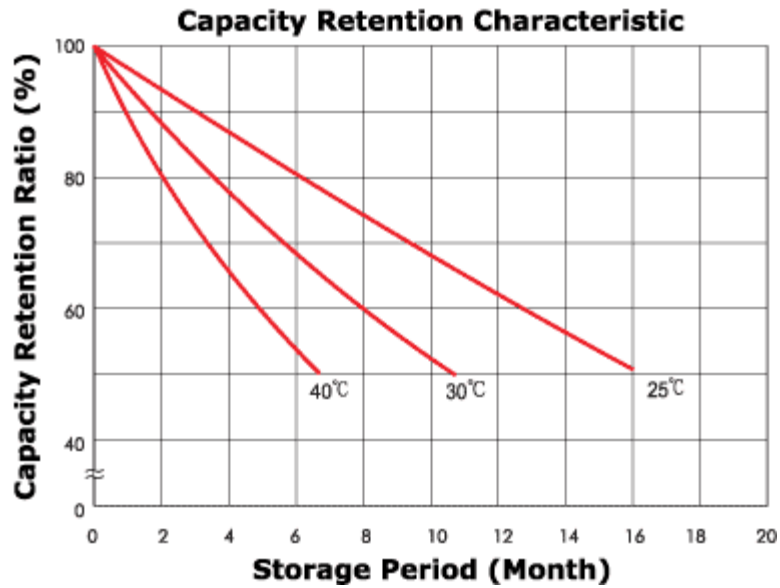


Chart 3

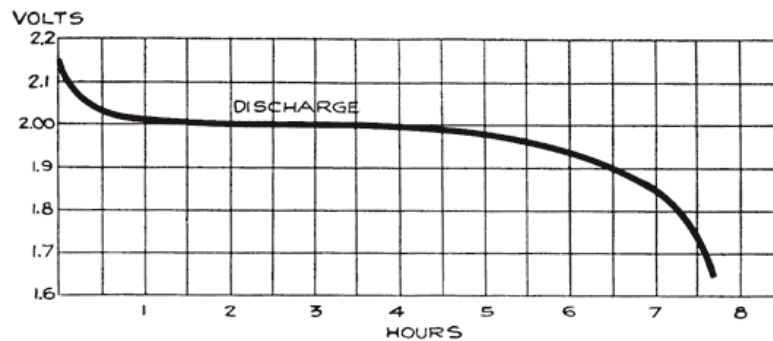


- Self-discharge (disposable battery) – 8 to 10 % loss of original charge
- Cold temperatures will slow these reactions down and warm temperatures will increase these reactions
- Memory effect- NiCd cells, if used in a particular repetitive manner, may show a decrease in capacity.

Environmental factors

- Leakage, gassing, toxicity
- Shock resistance

Discharge characteristics (voltage drop)



Battery self-discharge is caused by the internal reactions in a battery that reduce the energy stored without any connection with an external circuit. In other words, the battery loses the energy stored in it by itself due to its internal behaviour even when the connected application is not demanding

any energy.

Problem

2500 mAh it means that the battery has a capability to deliver 2.5A/2500mA of current to the load for 1 hour. The time that the battery works continuously depends upon the load current that it consumes. So if the load consumes only 25 mA of current then the battery can stay alive for 100 hours. How is it?

- If the load draws 2500 mA of current, an hour.
- If the load draws 250 mA of current, 10 hours.
- If the load draws 25 mA of current, 100 hours.

Convert mAh (milliamphours) to time in hours

The formula is (mAh)/(Amps*1000) = (hours).

For example, if you have a 3000 mAh battery that runs at 0.2 Amps (0.2Amps = 200mA), then the time that the battery will last for is $(3000)/(0.2*1000) = (3000)/(200) = 15$ Hours.

<https://milliamps-watts.appspot.com/>

<https://convert-formula.com/mah-hours>

Reversible cells and Irreversible cells

1. Reversible cells

Cells which obey the conditions of thermodynamic reversibility are called as reversible cells. Examples: Daniel cell, Secondary batteries (rechargeable batteries)

Danniel cell is an example of reversible cell. Its emf is 1.1 volt. It is represented as:



It is considered as a reversible cell , because it satisfies the following three conditions:

- If the danniel cell is connected to an external source of emf equal to 1.1 volt, no current flows and no chemical reaction takes place in the cell.
- If the external emf is made slightly less than 1.1 volt, small current flows from the cell and small chemical reaction occurs.
- If the external emf is made slightly higher than 1.1 volt, the current will flow in the opposite direction. Copper will pass into solution as copper ions and zinc will get deposited on the zinc electrode. In other words, reverse chemical reactions takes place when the current flows in the opposite direction.

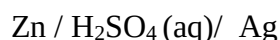
Since the secondary batteries (Lead acid battery, Nickel-Cadmium battery) obey these conditions, they can be recharged and are called rechargeable batteries.

2. Irreversible cells

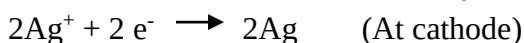
Cell which do not obey the conditions of thermodynamic reversibility are called irreversible cells.

Examples: Zinc-silver cell, Dry cell (primary cells)

Zinc-silver cell is an example for a irreversible cell. It is represented as



The reactions taking place at the anode and cathode are:



When two electrodes are connected from outside, zinc dissolves with the liberation of hydrogen gas.

When the external emf is slightly greater than the actual emf of the cell is applied to it, the above reaction are not reversed, since one the product, H_2 gas already escaped. Such a cell, which does not obey the conditions of thermodynamic reversibility is called an irreversible cell. Similarly primary cells (Eg. dry cell) do not obey the conditions of reversibility and hence they are not chargeable.

Types of batteries:

- Primary batteries (or) disposable batteries
- Secondary batteries (or) rechargeable batteries

Primary (Disposable) Batteries:

In these cells, the cell reactions occur only once. The electrodes and the electrode reactions cannot be reversed by passing an external electrical energy through them. They are not chargeable. Primary batteries have to be discarded after the exhaustion of their reactants. Examples: Dry cells, mercury cells.

Secondary batteries:

A rechargeable battery (storage battery, secondary cell, or accumulator) is a type of electrical battery which can be charged, discharged into a load, and recharged many times.

Differences between Primary and Secondary batteries:

Primary battery	Secondary battery
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1. These are non-rechargeable and meant for a single use and to be discarded after use. 2. Cell reaction is not reversible. 3. Cannot be rechargeable. 4. Less expensive. 5. Can be used as long as the materials are active in their composition. Eg: Leclanche cell, Dry Cells, Mercury Cells	1. These are rechargeable and meant for multi cycle use. 2. Cell reaction can be reversed. 3. Can be rechargeable. 4. Expensive. 5. Can be used again and again by recharging the cell. Eg; Lead- acid cell, Ni-cd cells
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